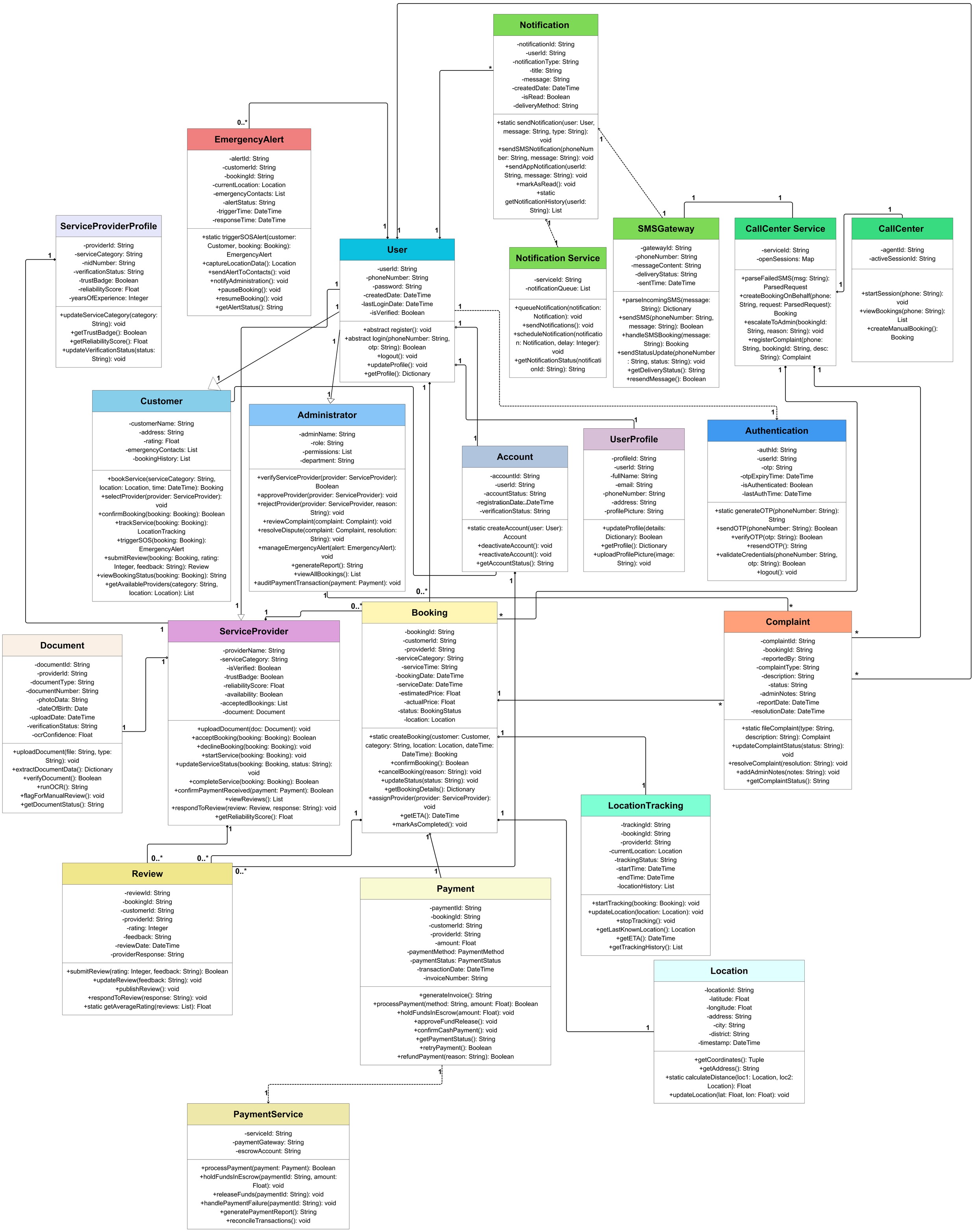
Notification
-notificationId: String
-userId: String
-notificationType: String
-title: String
-message: String
-createdDate: DateTime
-isRead: Boolean
-deliveryMethod: String
+static sendNotification(user: User, message: String, type: String): void
+sendSMSNotification(phoneNumber: String, message: String): void
+sendAppNotification(userId: String, message: String): void
+markAsRead(): void
+static getNotificationHistory(userId: String): List
EmergencyAlert
-alertId: String
-customerId: String
-bookingId: String
-currentLocation: Location
-emergencyContacts: List
-alertStatus: String
-triggerTime: DateTime
-responseTime: DateTime
+static triggerSOSAlert(customer: Customer, booking: Booking): EmergencyAlert
+captureLocationData(): Location
+sendAlertToContacts(): void
+notifyAdministration(): void
+pauseBooking(): void
+resumeBooking(): void
+getAlertStatus(): String
ServiceProviderProfile
-providerId: String
-serviceCategory: String
-nidNumber: String
-verificationStatus: String
-trustBadge: Boolean
-reliabilityScore: Float
-yearsOfExperience: Integer
+updateServiceCategory(category: String): void
+getTrustBadge(): Boolean
+getReliabilityScore(): Float
+updateVerificationStatus(status: String): void
User
-userId: String
-phoneNumber: String
-password: String
-createdDate: DateTime
-lastLoginDate: DateTime
-isVerified: Boolean
+abstract register(): void
+abstract login(phoneNumber: String, otp: String): Boolean
+logout(): void
+updateProfile(): void
+getProfile(): Dictionary
Notification Service
-serviceId: String
-notificationQueue: List
+queueNotification(notification: Notification): void
+sendNotifications(): void
+scheduleNotification(notification: Notification, delay: Integer): void
+getNotificationStatus(notificationId: String): String
SMSGateway
-gatewayId: String
-phoneNumber: String
-messageContent: String
-deliveryStatus: String
-sentTime: DateTime
+parseIncomingSMS(message: String): Dictionary
+sendSMS(phoneNumber: String, message: String): Boolean
+handleSMSBooking(message: String): Booking
+sendStatusUpdate(phoneNumber: String, status: String): void
+getDeliveryStatus(): String
+resendMessage(): Boolean
CallCenter Service
-serviceId: String
-openSessions: Map
+parseFailedSMS(msg: String): ParsedRequest
+createBookingOnBehalf(phone: String, request: ParsedRequest): Booking
+escalateToAdmin(bookingId: String, reason: String): void
+registerComplaint(phone: String, bookingId: String, desc: String): Complaint
CallCenter
-agentId: String
-activeSessionId: String
+startSession(phone: String): void
+viewBookings(phone: String): List
+createManualBooking(): Booking
Customer
-customerName: String
-address: String
-rating: Float
-emergencyContacts: List
-bookingHistory: List
+bookService(serviceCategory: String, location: Location, time: DateTime): Booking
+selectProvider(provider: ServiceProvider): void
+confirmBooking(booking: Booking): Boolean
+trackService(booking: Booking): LocationTracking
+triggerSOS(booking: Booking): EmergencyAlert
+submitReview(booking: Booking, rating: Integer, feedback: String): Review
+viewBookingStatus(booking: Booking): String
+getAvailableProviders(category: String, location: Location): List
Administrator
-adminName: String
-role: String
-permissions: List
-department: String
+verifyServiceProvider(provider: ServiceProvider): Boolean
+approveProvider(provider: ServiceProvider): void
+rejectProvider(provider: ServiceProvider, reason: String): void
+reviewComplaint(complaint: Complaint): void
+resolveDispute(complaint: Complaint, resolution: String): void
+manageEmergencyAlert(alert: EmergencyAlert): void
+generateReport(): String
+viewAllBookings(): List
+auditPaymentTransaction(payment: Payment): void
Account
-accountId: String
-userId: String
-accountStatus: String
-registrationDate: DateTime
-verificationStatus: String
+static createAccount(user: User): Account
+deactivateAccount(): void
+reactivateAccount(): void
+getAccountStatus(): String
UserProfile
-profileId: String
-userId: String
-fullName: String
-email: String
-phoneNumber: String
-address: String
-profilePicture: String
+updateProfile(details: Dictionary): Boolean
+getProfile(): Dictionary
+uploadProfilePicture(image: String): void
Authentication
-authId: String
-userId: String
-otp: String
-otpExpiryTime: DateTime
-isAuthenticated: Boolean
-lastAuthTime: DateTime
+static generateOTP(phoneNumber: String): String
+sendOTP(phoneNumber: String): Boolean
+verifyOTP(otp: String): Boolean
+resendOTP(): String
+validateCredentials(phoneNumber: String, otp: String): Boolean
+logout(): void
Document
-documentId: String
-providerId: String
-documentType: String
-documentNumber: String
-photoData: String
-dateOfBirth: Date
-uploadDate: DateTime
-verificationStatus: String
-ocrConfidence: Float
+uploadDocument(file: String, type: String): void
+extractDocumentData(): Dictionary
+verifyDocument(): Boolean
+runOCR(): String
+flagForManualReview(): void
+getDocumentStatus(): String
ServiceProvider
-providerName: String
-serviceCategory: String
-isVerified: Boolean
-trustBadge: Boolean
-reliabilityScore: Float
-availability: Boolean
-acceptedBookings: List
-document: Document
+uploadDocument(doc: Document): void
+acceptBooking(booking: Booking): Boolean
+declineBooking(booking: Booking): void
+startService(booking: Booking): void
+updateServiceStatus(booking: Booking, status: String): void
+completeService(booking: Booking): Boolean
+confirmPaymentReceived(payment: Payment): Boolean
+viewReviews(): List
+respondToReview(review: Review, response: String): void
+getReliabilityScore(): Float
Booking
-bookingId: String
-customerId: String
-providerId: String
-serviceCategory: String
-serviceTime: String
-bookingDate: DateTime
-serviceDate: DateTime
-estimatedPrice: Float
-actualPrice: Float
-status: BookingStatus
-location: Location
+static createBooking(customer: Customer, category: String, location: Location, dateTime: DateTime): Booking
+confirmBooking(): Boolean
+cancelBooking(reason: String): void
+updateStatus(status: String): void
+getBookingDetails(): Dictionary
+assignProvider(provider: ServiceProvider): void
+getETA(): DateTime
+markAsCompleted(): void
Complaint
-complaintId: String
-bookingId: String
-reportedBy: String
-complaintType: String
-description: String
-status: String
-adminNotes: String
-reportDate: DateTime
-resolutionDate: DateTime
+static fileComplaint(type: String, description: String): Complaint
+updateComplaintStatus(status: String): void
+resolveComplaint(resolution: String): void
+addAdminNotes(notes: String): void
+getComplaintStatus(): String
LocationTracking
-trackingId: String
-bookingId: String
-providerId: String
-currentLocation: Location
-trackingStatus: String
-startTime: DateTime
-endTime: DateTime
-locationHistory: List
+startTracking(booking: Booking): void
+updateLocation(location: Location): void
+stopTracking(): void
+getLastKnownLocation(): Location
+getETA(): DateTime
+getTrackingHistory(): List
Review
-reviewId: String
-bookingId: String
-customerId: String
-providerId: String
-rating: Integer
-feedback: String
-reviewDate: DateTime
-providerResponse: String
+submitReview(rating: Integer, feedback: String): Boolean
+updateReview(feedback: String): void
+publishReview(): void
+respondToReview(response: String): void
+static getAverageRating(reviews: List): Float
Payment
-paymentId: String
-bookingId: String
-customerId: String
-providerId: String
-amount: Float
-paymentMethod: PaymentMethod
-paymentStatus: PaymentStatus
-transactionDate: DateTime
-invoiceNumber: String
+generateInvoice(): String
+processPayment(method: String, amount: Float): Boolean
+holdFundsInEscrow(amount: Float): void
+approveFundRelease(): void
+confirmCashPayment(): void
+getPaymentStatus(): String
+retryPayment(): Boolean
+refundPayment(reason: String): Boolean
Location
-locationId: String
-latitude: Float
-longitude: Float
-address: String
-city: String
-district: String
-timestamp: DateTime
+getCoordinates(): Tuple
+getAddress(): String
+static calculateDistance(loc1: Location, loc2: Location): Float
+updateLocation(lat: Float, lon: Float): void
PaymentService
-serviceId: String
-paymentGateway: String
-escrowAccount: String
+processPayment(payment: Payment): Boolean
+holdFundsInEscrow(paymentId: String, amount: Float): void
+releaseFunds(paymentId: String): void
+handlePaymentFailure(paymentId: String): void
+generatePaymentReport(): String
+reconcileTransactions(): void